PRACTICAL ASSIGNMENT REPORT

NPC – UNIT 3

CONFIGURE A DNS SERVER AND WEB SERVER USING CISCO PACKET TRACER

Submitted By

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GitHub Repository: github.com/Pranavkotame/Unit_3_DNS_Web_Server

A practical report submitted towards the Networking Protocols & Communication (NPC) course, demonstrating the configuration and testing of DNS name resolution and HTTP web access within a local area network built in Cisco Packet Tracer.

1. Aim and Objectives

Aim

To configure a DNS Server and a Web Server using Cisco Packet Tracer and to access a website through its domain name instead of its IP address.

Objectives

1. Create a network topology using PCs, a switch, a DNS Server and a Web Server.
2. Assign appropriate IP addresses to all devices.
3. Configure and enable the HTTP service on the Web Server.
4. Configure the DNS service and create an A (Address) record.
5. Configure the PCs to use the DNS Server for name resolution.
6. Verify connectivity between devices using ping commands.
7. Access the hosted website using its domain name from a client PC.

2. Requirements

The following devices and software were used to carry out this practical. The role of each component is described briefly.

- **Cisco Packet Tracer** – network simulation software used to design, configure and test the topology virtually.
- **PC0 and PC1 (PC-PT)** – client end devices that send DNS queries and request web pages from the server.
- **Cisco 2960 Switch (Switch0)** – a Layer-2 device that connects all end devices within the same LAN and forwards frames based on MAC addresses.
- **DNS Server (Server-PT)** – resolves the domain name to the Web Server's IP address by hosting the DNS A record.
- **Web Server (Server-PT)** – hosts the website and serves the web page to clients over the HTTP protocol.
- **Copper Straight-Through cables** – connect the end devices and servers to the switch (used between devices of different types).

3. Network Topology

The network is a single local area network (LAN) belonging to the subnet 192.168.1.0/24. The DNS Server, Web Server, PC0 and PC1 are each connected to the central switch, Switch0, using Copper Straight-Through cables. Because every device is in the same subnet, communication takes place through the switch without the need for a router.

Logical layout

```
DNS Server --+
Web Server --+
PC0 ---------+--- Switch0
PC1 ---------+
```

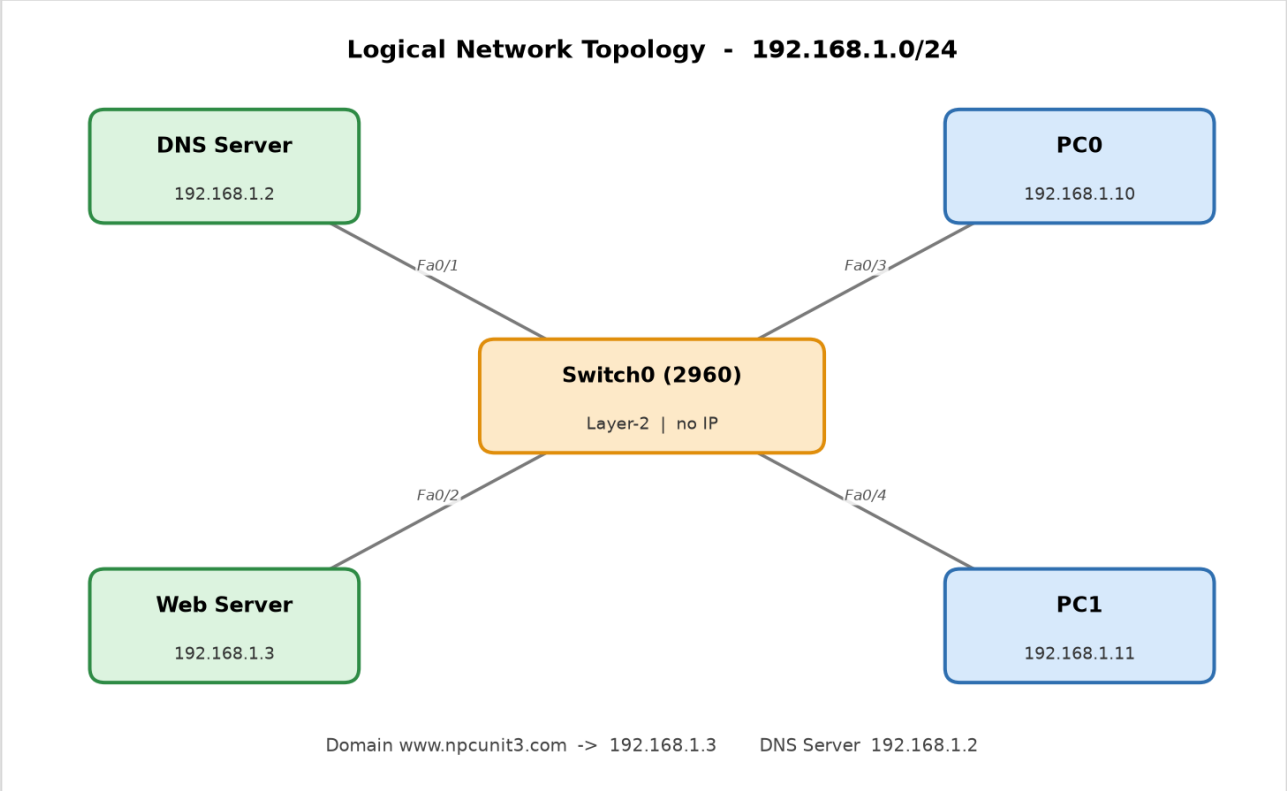


Figure 3.1: Logical network topology (reference design)

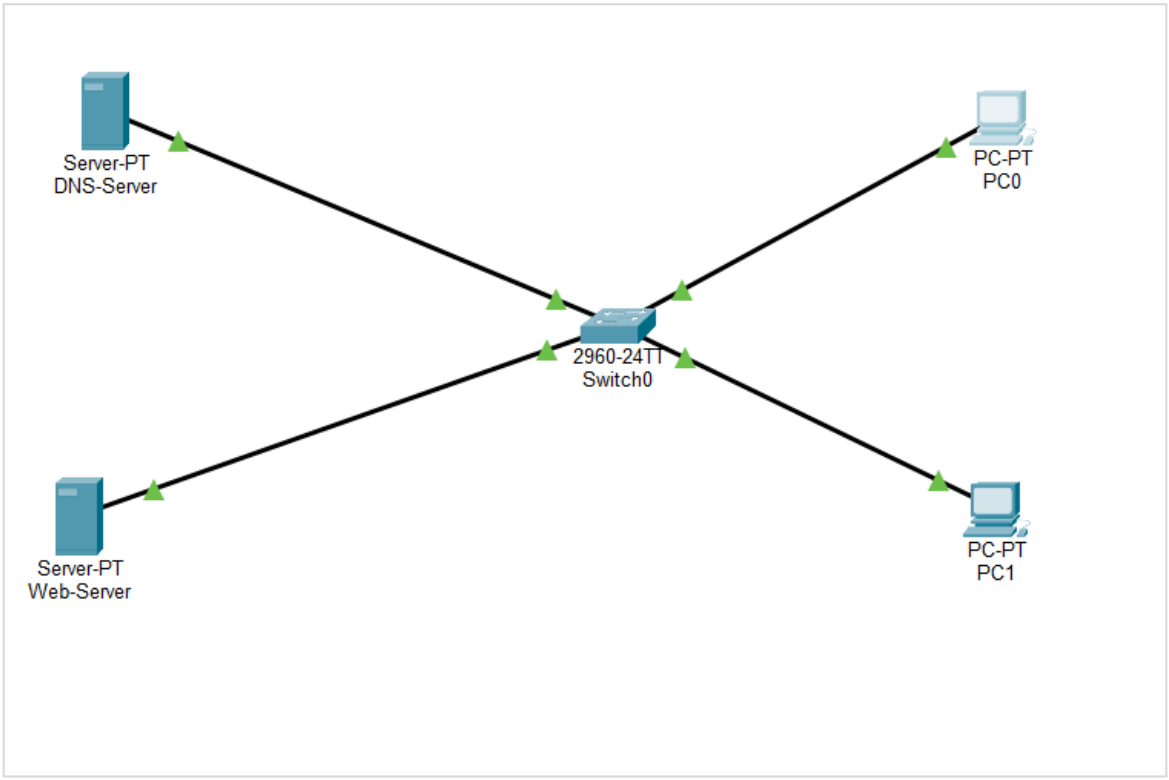


Figure 3.2: Topology as implemented in Cisco Packet Tracer

4. IP Addressing Table

Static IP addressing was used for all devices. The addressing scheme is shown in the table below.

Device	IP Address	Subnet Mask
DNS Server	192.168.1.2	255.255.255.0
Web Server	192.168.1.3	255.255.255.0
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0

DNS Server address configured on the PCs	192.168.1.2
Default gateway	Not required for communication within this single subnet.

5. DNS Server Configuration

1. Open the DNS Server.
2. Navigate to Desktop > IP Configuration.
3. Assign the IP address 192.168.1.2.
4. Set the subnet mask to 255.255.255.0.
5. Navigate to Services > DNS.
6. Turn the DNS service ON.
7. Create the DNS resource record shown below.

Name	Type	Address
www.npcunit3.com	A Record	192.168.1.3

The DNS A (Address) record maps the human-readable domain name **www.npcunit3.com** to the Web Server's IPv4 address **192.168.1.3**. When a PC requests the domain name, the DNS Server returns this address so the browser can reach the Web Server.

6. Web Server Configuration

1. Open the Web Server.
2. Navigate to Desktop > IP Configuration.
3. Assign the IP address 192.168.1.3.
4. Set the subnet mask to 255.255.255.0.
5. Navigate to Services > HTTP.
6. Enable the HTTP service.
7. Edit index.html to display a simple welcome page.

index.html

```
<html>
<head>
    <title>NPC Unit 3</title>
</head>
<body>
    <h1>Welcome to My Web Server!</h1>
    <p>DNS and Web Server Configuration Successful.</p>
</body>
</html>
```

The HTML page is stored on the Web Server and delivered to the client through the HTTP service whenever the website is requested.

7. PC Configuration

PC0 and PC1 were configured with static IP addresses through Desktop > IP Configuration. The DNS Server field on each PC points to the DNS Server at 192.168.1.2.

PC0

IP Address	192.168.1.10
Subnet Mask	255.255.255.0
DNS Server	192.168.1.2

PC1

IP Address	192.168.1.11
Subnet Mask	255.255.255.0
DNS Server	192.168.1.2

Static addressing was used throughout this practical so that the addresses remain fixed and predictable.

8. Connectivity Testing

Connectivity was verified from PC0 using the ping command. The following commands were executed in Desktop > Command Prompt:

```
C:\>ping 192.168.1.2      (DNS Server)
C:\>ping 192.168.1.3      (Web Server)
```

Observed result: both ping tests returned 4 of 4 replies with 0% packet loss (time < 1 ms, TTL = 128), which confirms successful IP connectivity between PC0 and both servers. The screenshot below shows the actual output.

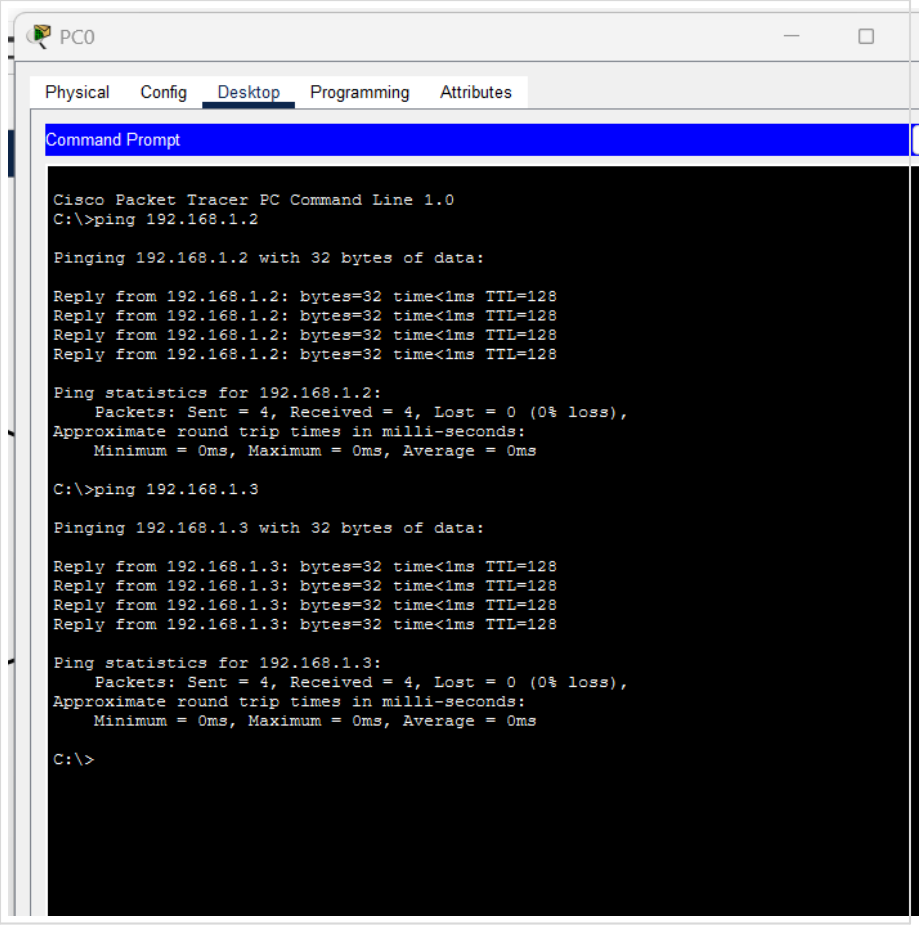


Figure 8.1: Ping test to the DNS Server (192.168.1.2) and Web Server (192.168.1.3) from PC0

9. Website Access and Result

The website was accessed from a client PC using its domain name:

1. Open PC0.
2. Navigate to Desktop > Web Browser.
3. Enter the URL http://www.npcunit3.com
4. Click Go.

Expected / observed web page output:

```
Welcome to My Web Server!
DNS and Web Server Configuration Successful.
```

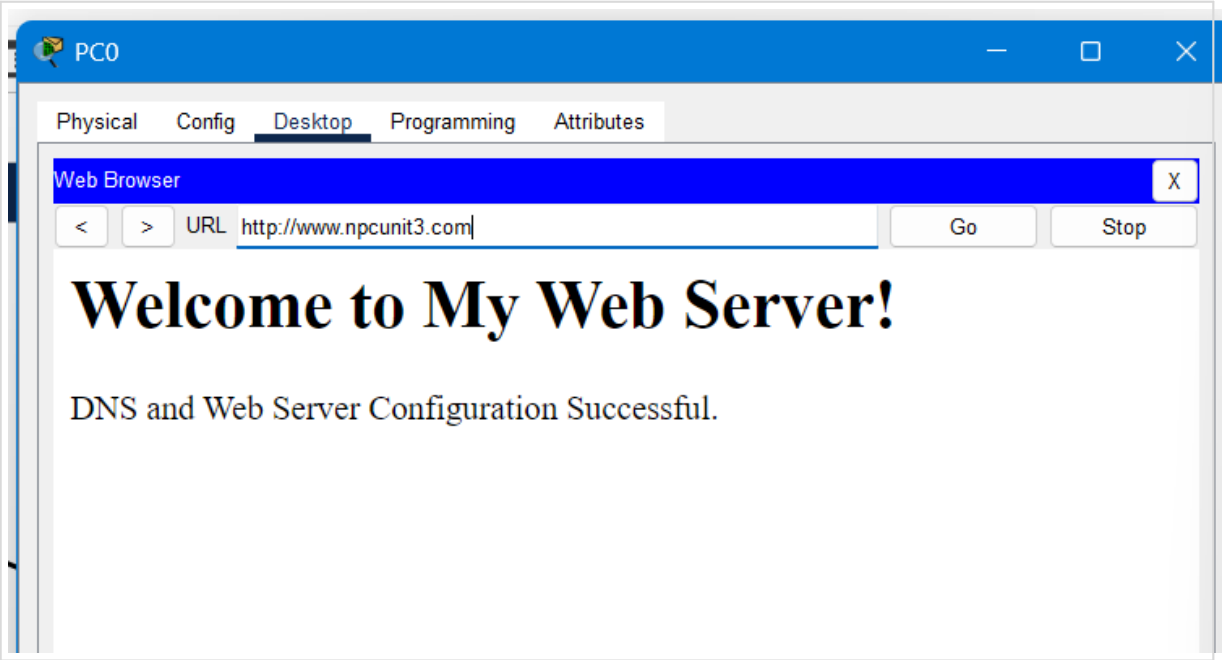


Figure 9.1: Website opened on PC0 through the domain name http://www.npcunit3.com

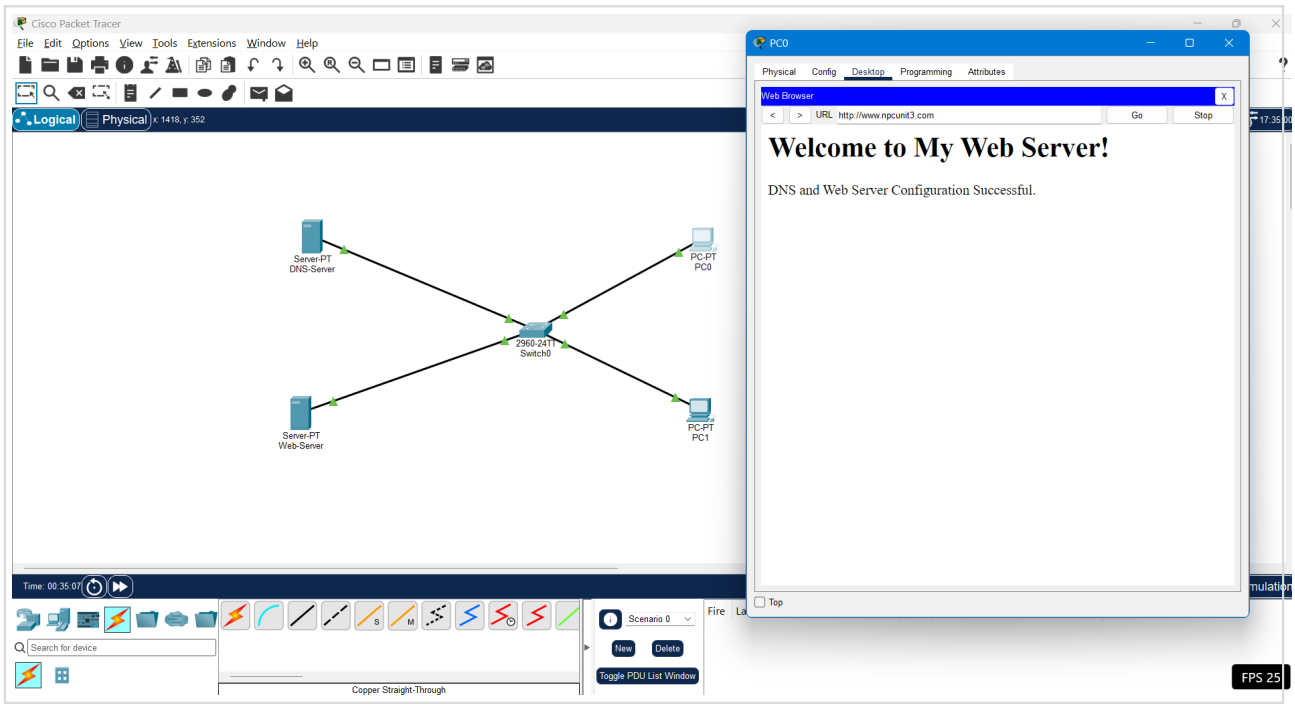


Figure 9.2: Complete Packet Tracer workspace with the website served to PC0

Successful access to the website through the domain name demonstrates that DNS name resolution and HTTP web access are both working correctly: the domain name was resolved to 192.168.1.3 by the DNS Server, and the page was then served by the Web Server over HTTP.

10. Troubleshooting

Common problems that may occur during this practical and their solutions are listed below.

Problem	Solution
Website domain does not resolve	Check that the DNS service is ON and the A record for www.npcunit3.com (192.168.1.3) exists.
Website cannot be accessed	Check that the HTTP service is enabled and the Web Server IP (192.168.1.3) is configured correctly.
Ping fails	Verify IP addresses, subnet masks, cables and switch connections (link lights green).
Incorrect DNS server address	Configure 192.168.1.2 in the DNS Server field of each PC.
Incorrect domain name	Ensure the browser URL exactly matches the DNS record (www.npcunit3.com).

11. Conclusion

In this practical, a local area network was successfully designed and configured in Cisco Packet Tracer using two PCs, a switch, a DNS Server and a Web Server, all addressed within the 192.168.1.0/24 subnet. The HTTP service was enabled on the Web Server to host a simple web page, and a DNS A record was created so that the domain name **www.npcunit3.com** maps to the Web Server's IP address **192.168.1.3**.

Connectivity was verified with the ping command, and the website was accessed from PC0 through its domain name rather than its IP address. This confirms that DNS name resolution and HTTP website access function correctly within the local network, demonstrating how clients, a DNS Server and a Web Server interact in a TCP/IP network.